

Example sub-graph

For "Buffer overflow in OpenSSL 1.0.1f"

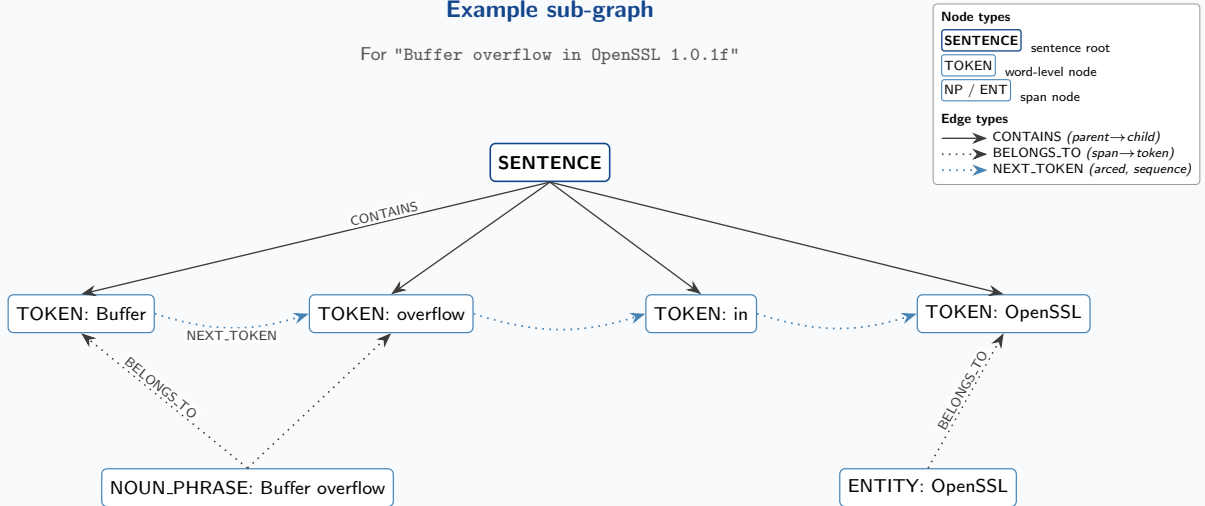


Figure. A small piece of the Text Property Graph we build from the CVE description "Buffer overflow in OpenSSL 1.0.1f". spaCy turns every word into a TOKEN node and links neighbouring tokens with NEXT_TOKEN edges in reading order. The SENTENCE node sits at the top and reaches every TOKEN through CONTAINS edges. Two multi-word spans appear at the bottom: the noun phrase "Buffer overflow" and the named entity "OpenSSL". Each one points to the tokens it covers via BELONGS_TO. SENTENCE also has direct CONTAINS edges to NOUN_PHRASE and ENTITY in the real graph, but we leave those out of the figure because a straight line would cut through TOKEN: overflow and TOKEN: in; the same containment is still recoverable through the remaining CONTAINS + BELONGS_TO chain.